

Chapter 4

1. Parity Conditions in International Finance and Currency Forecasting

Chapter Outline

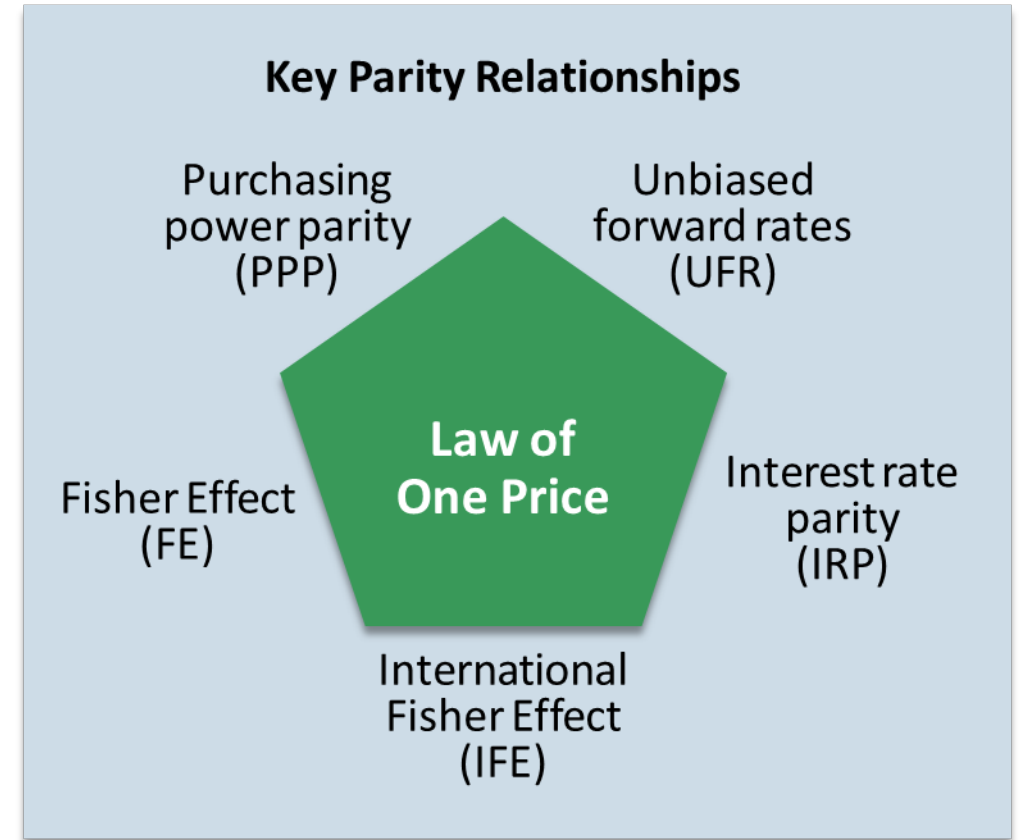
- Parity conditions
- Purchasing power parity
- Fisher effect
- International Fisher effect
- Interest rate parity
- Unbiased forward rates
- Currency forecasting

Parity Conditions

- What are parity conditions?
- Law of one price example

What are Parity Conditions?

- Parity conditions
 - Relationships based on the *Law of One Price* that are *ideally* enforced by arbitrage.
 - **Law of One Price** – in competitive markets, exchange-adjusted prices of identical tradable goods and financial assets must be within transaction costs of equality worldwide.
 - **Arbitrage** – the simultaneous purchase and sale of the same assets or commodities on different markets to profit from price discrepancies.
 - Absent market imperfections, arbitrage ensures that exchange-adjusted prices of identical traded goods follow the Law of One Price.



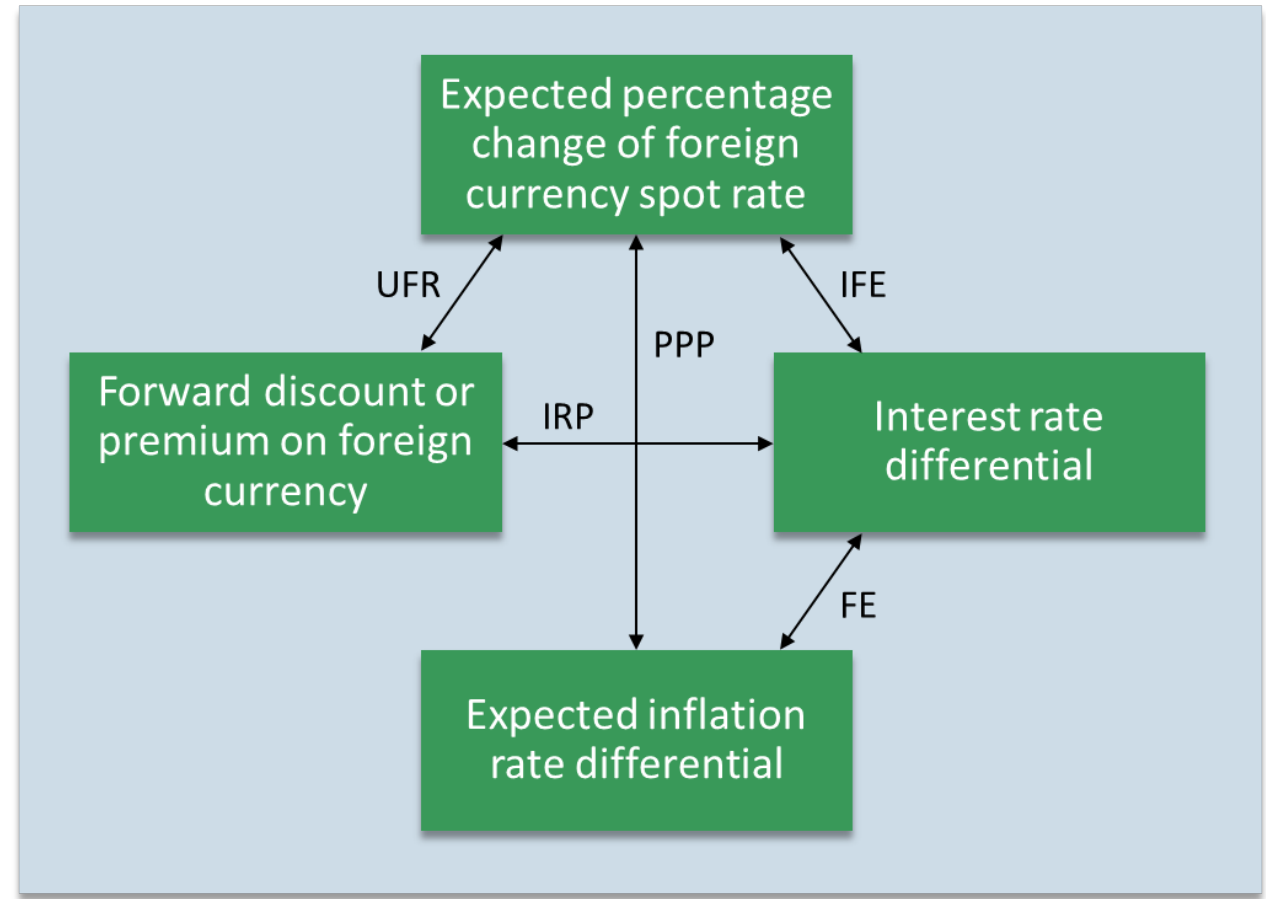
What are Parity Conditions?

Purchasing power parity	PPP	For prices in two countries to be equal, the exchange rate between the two countries must change by the difference between the domestic and foreign rates of inflation.
Fisher effect	FE	If expected real interest rates differ between the home and foreign countries, capital will flow to the country with the higher real rate until the real rates in both countries are equal and equilibrium is reached.
International Fisher effect	IFE	Combines the conditions underlying PPP and FE: If real interest rates differ between the home and foreign countries, capital will flow to the country with the higher real rate until the exchange-adjusted returns are equal in both countries and equilibrium is reached.
Interest rate parity	IRP	In an efficient market with no transaction costs, the interest rate differential between two countries should approximate the forward differential.
Unbiased forward rates	UFR	Equilibrium is achieved when the forward differential equals the expected change in the spot rate.

What are Parity Conditions?

- Parity relationships link various international factors together:
 - Spot exchange rate,
 - Forward exchange rate,
 - Interest rates, and
 - Real markets.

Parity Conditions	
Purchasing power parity	PPP
Fisher effect	FE
International Fisher effect	IFE
Interest rate parity	IRP
Unbiased forward rates	UFR



Expected percentage
change of foreign
currency spot rate

UFR

Forward discount or
premium on foreign
currency

IRP

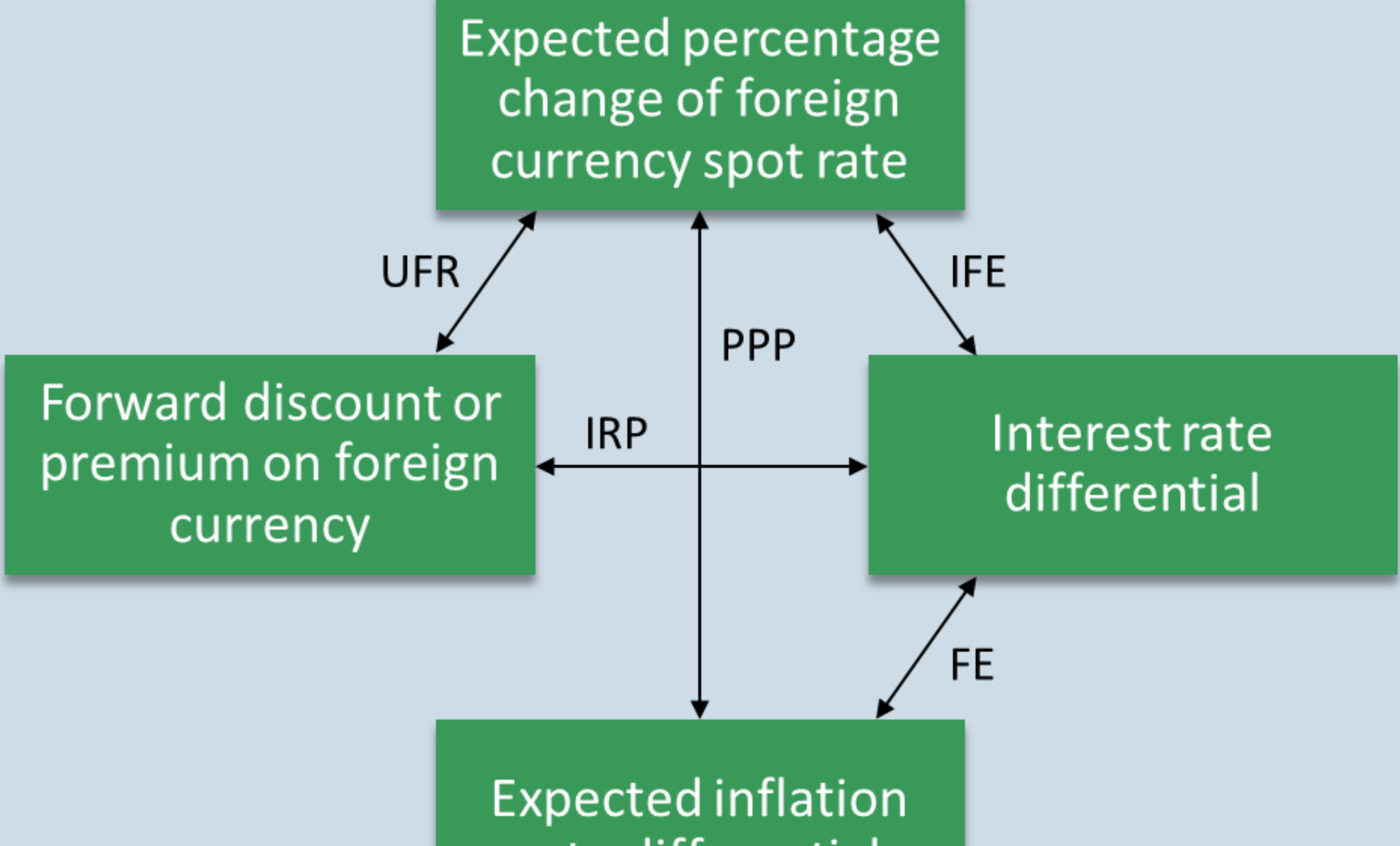
PPP

IFE

Interest rate
differential

FE

Expected inflation
rate differential



Law of One Price Example

- Novo Nordisk A/S, a leading Denmark-based pharmaceutical firm specializing in insulin production and delivery.
- It trades on the Copenhagen stock exchange under NOVO and on the New York stock exchange under NVO.
 - One share of NOVO = DKK 441.65
 - The exchange rate is DKK 6.89 /USD.
 - What should one share of NVO trade at in USD if the law of one price holds? $\Rightarrow \frac{441.65}{6.89} = \64.10
- What if NVO is worth \$65?
 - Arbitrageurs will short NVO and buy NOVO until the prices are equal.

Purchasing Power Parity (PPP)

- Two versions of PPP
- Absolute PPP
- Relative PPP
- Relative PPP intuitive approximation
- Monetary approach to PPP
- Empirical evidence on PPP

Two Versions of PPP

Absolute PPP

Asks, “What should the current exchange rate be?”

(i.e., price *level* e_0)

Relative PPP

Asks, “How should the exchange rate e_0 change?”

(i.e., $e_1 - e_0 = \Delta e$)

Absolute PPP

- A particular good must have the same price in different countries when priced in a common currency.
- With free trade, prices will equalize; otherwise arbitrage opportunities will exist.

Price Parity

$$P_h = e_0 P_f$$

Where

P_h = price of a good in terms of home currency (\$)

P_f = price of a good in foreign country

Using price parity, we can solve for any variable:

$$e_0 = \frac{P_h}{P_f}$$

Absolute PPP

Example: *The Economist* Big Mac Index

- In May 2004
 - Cost of Big Mac in US = \$3.00
 - Cost of Big Mac in South Africa = R12.41
- In January 2024:
 - Cost of Big Mac in US = \$5.69
 - Cost of Big Mac in South Africa = R51.90

According to PPP: $e_{ppp} = \frac{P_{us}}{P_{sa}} = \frac{\$3.00}{R12.41} = \$0.2417$

Actual: $e_{actual} = \$0.1476$ $\neq$

According to PPP, in 2004, the South African rand was undervalued.

According to PPP: $e_{ppp} = \frac{P_{us}}{P_{sa}} = \frac{\$5.69}{R51.90} = \$0.1096$

Actual: $e_{actual} = \$0.0521$ $\neq$

According to PPP, in 2024, the South African rand was undervalued.

Absolute PPP

Example

- Suppose a Jeep Grand Cherokee costs:
 - \$35,000 in the US
 - £25,000 in the UK
- Suppose the spot exchange rate is £1 = \$1.60.
- Does PPP hold?

According to PPP: $e_{ppp} = \frac{P_{us}}{P_{uk}} = \frac{\$35,000}{£25,000} = \$1.40$

Actual: $e_{actual} = \$1.60$ $\neq$



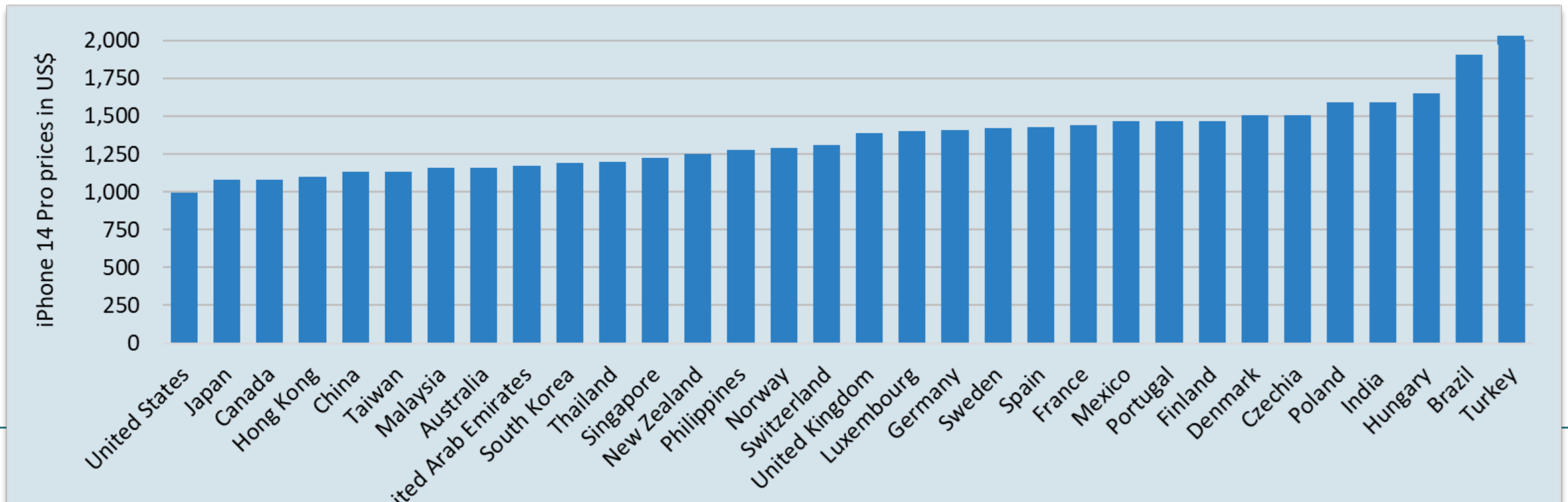
Or, another way: $£25,000 \times 1.6 = \$40,000, \neq \$35,000$

PPP does not hold.

If it did, $P_{uk} \times \$1.60$ would equal \$35,000, not \$40,000

Empirical Evidence on absolute PPP

- Absolute PPP does not hold:
 - Risk and costs of shipping goods internationally, tariffs, and other trade barriers prevent the law of one price from taking hold.
 - E.g., an iPhone 14 Pro is \$995 in the U S and \$2,010 in Turkey.



Relative PPP

- Exchange rates will change to offset changes in inflation, thereby equalizing purchasing power.

Relative PPP

$$e_t = e_0 \times \frac{(1 + i_h)^t}{(1 + i_f)^t}$$

Where

e_t = expected spot exchange rate at future time t

e_0 = today's sport exchange rate

i_h = home country inflation rate

i_f = foreign country inflation rate

Relative PPP

Example

- Suppose:
 - $i_{\$} = 5\%$
 - $i_{¥} = 1\%$
- What happens?
 - US goods are more expensive
 - Consumers switch to Japanese goods
 - ¥ increases against the \$
- How much will yen increase?
 - Until the Japanese goods are no longer cheaper than in US dollar terms: roughly 4%.

Relative PPP

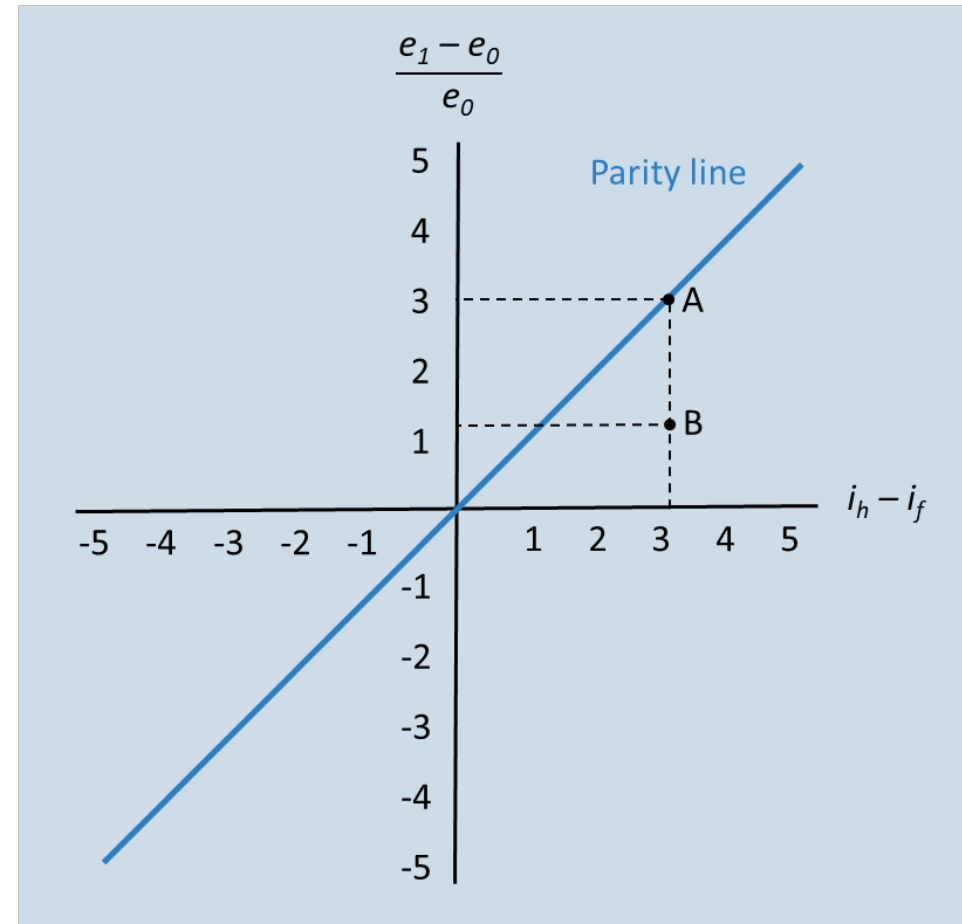
Example

- Suppose:
 - $i_{\$} = 5\%$ per annum
 - $i_{¥} = 1\%$ per annum
 - $e_0 = \$0.009447$
- If relative PPP holds, what should the $\$/¥$ exchange rate be in 5 years?

$$e_t = e_0 \times \frac{(1+i_h)^t}{(1+i_f)^t} \Rightarrow e_t = \$0.009447 \times \frac{(1.05)^5}{(1.01)^5} = \$0.011472$$

Relative PPP Intuitive Approximation

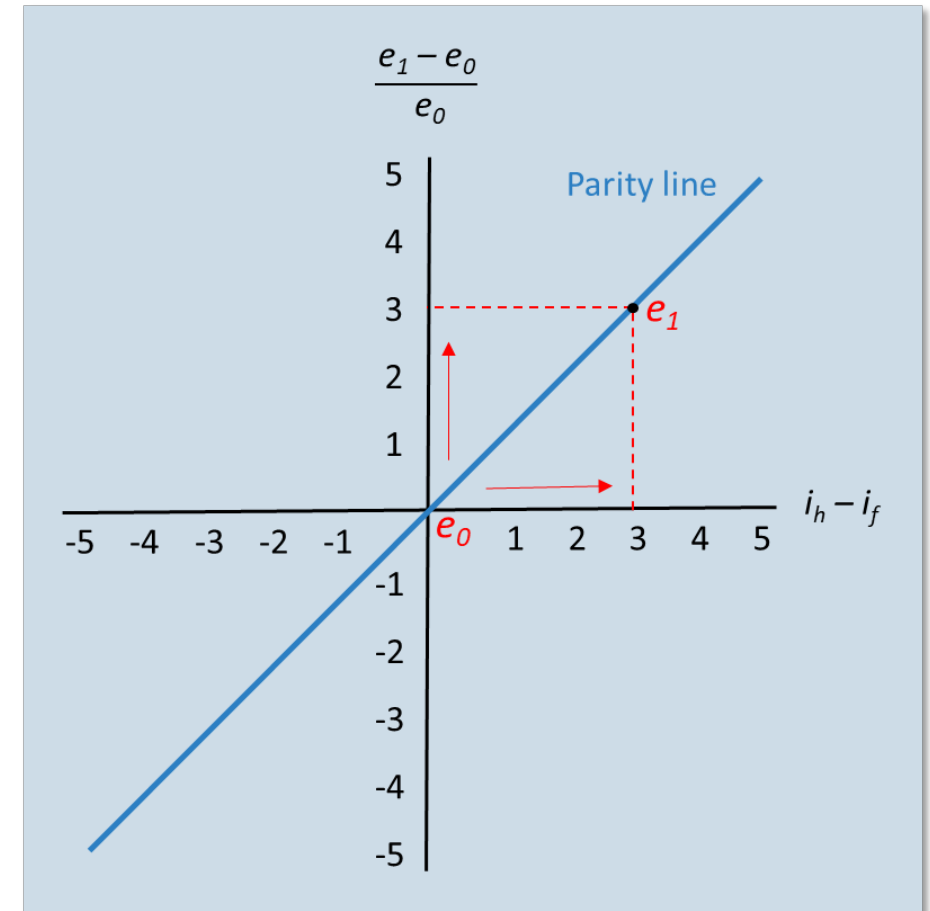
$$\underbrace{\frac{e_t - e_0}{e_0}}_{\text{\% change in exchange rate}} \approx \underbrace{i_h - i_f}_{\text{Difference in inflation rates}}$$



Relative PPP Intuitive Approximation

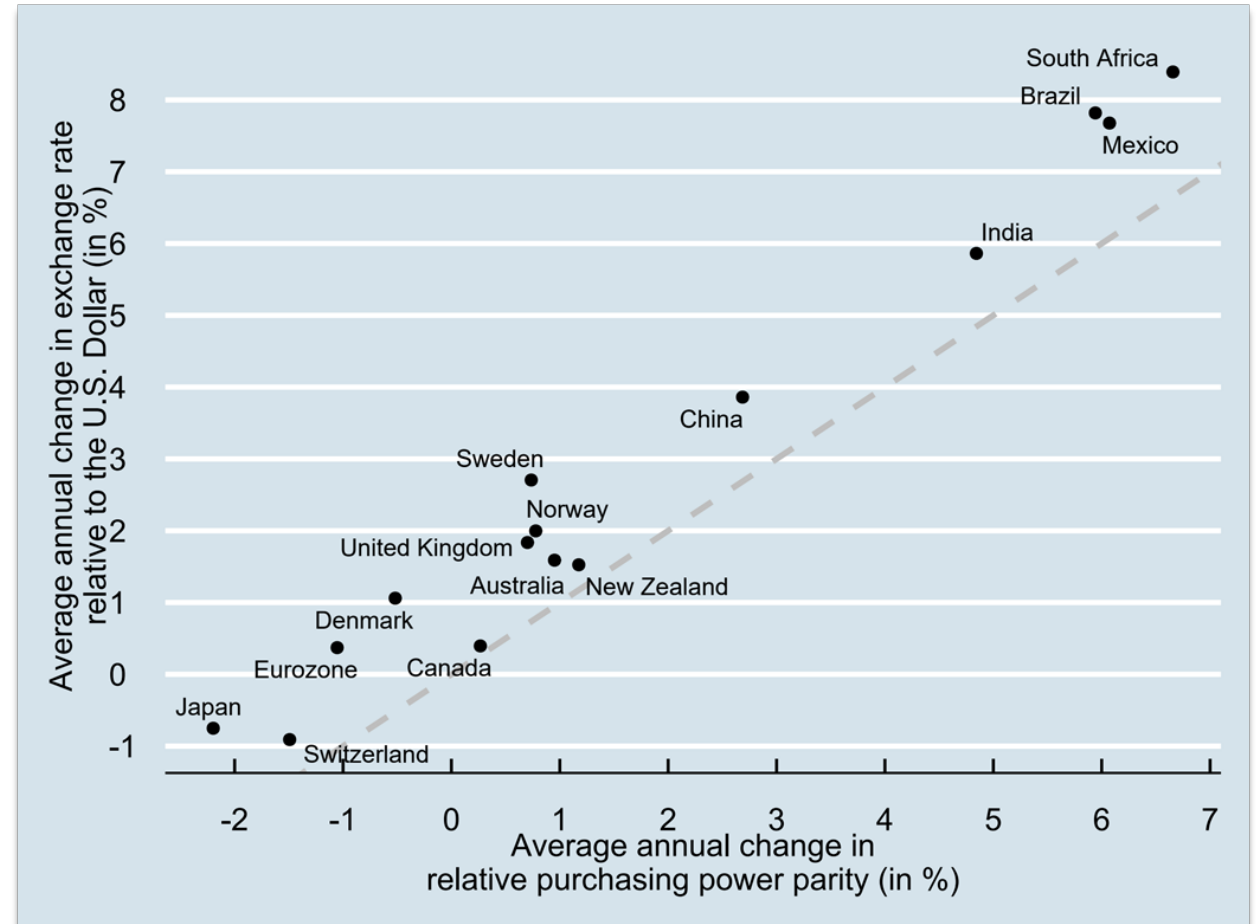
Example

1. At equilibrium e_0 , U.S. and Japan inflation both = 2% and change in $e_0 = 0$
2. U.S. inflation increases to 5% ($i_h - i_f = 3\%$), causing disequilibrium, or purchasing power disparity, at (0,3).
3. \$/¥ exchange rate will rise $\sim 3\%$ ($e_1 - e_0 / e_0 \approx 3\%$) to equalize the dollar price of goods in the two countries at a new equilibrium e_1 .



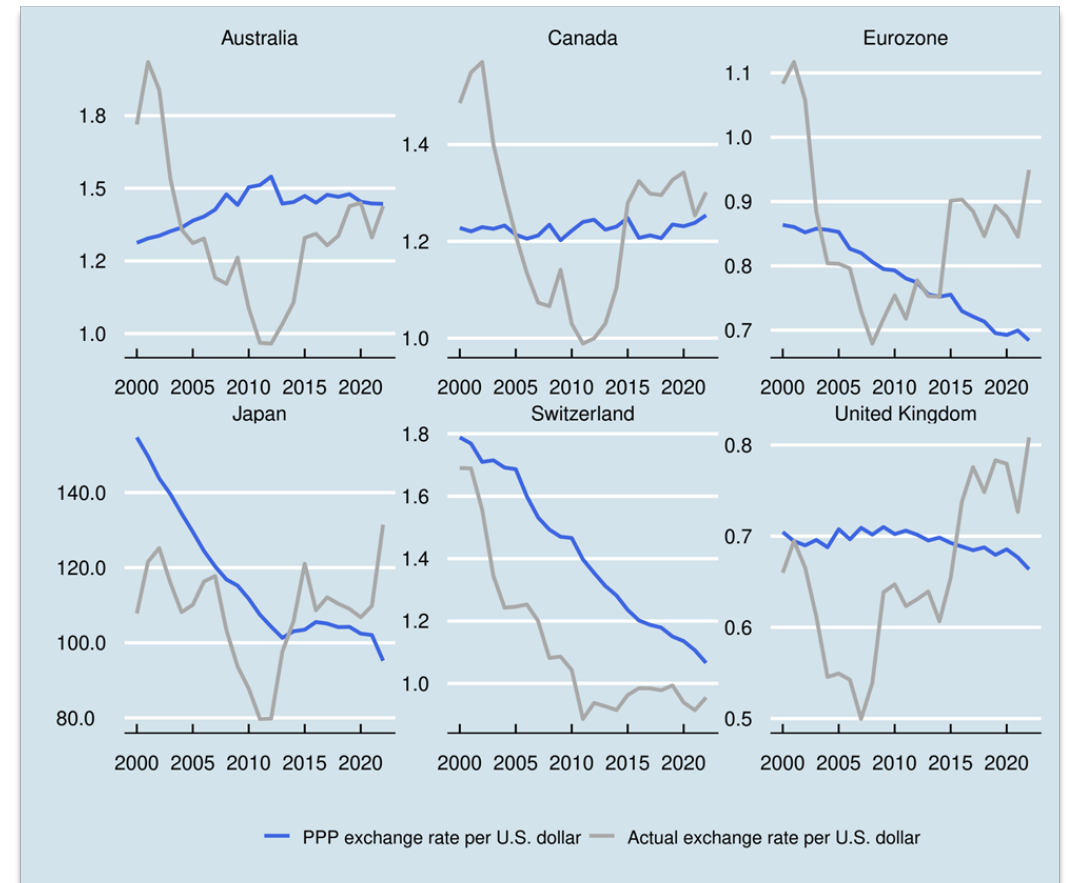
Empirical Evidence on Relative PPP

- Relative PPP does not hold in the short run.
 - Exchange rates change quickly.
 - Prices of goods are sticky. Exchange rates follow with a lag.
 - Differently constructed price indices and nontraded goods and services (e.g., haircuts).
- Relative PPP does hold in the long run.
 - There is a clear relationship between relative inflation rates and changes in exchange rates.



Empirical Evidence on PPP

- Changes in actual exchange rates and relative PPP for select countries.
- Despite substantial short-run deviations from purchasing power parity, currencies have a distinct tendency to move toward their PPP-predicted rates.
- However, as you can see from these graphs, there is a major difference in trend between the relative PPP and the actual exchange rate per US dollar.



The Fisher Effect (FE)

- The Fisher effect
- Generalized Fisher effect
- Intuitive approximation of the Fisher effect
- Empirical evidence for the Fisher effect

Fisher Effect

1. Fisher effect states that the nominal interest rate (r) is made up of two components:
2. A real required rate of return (a)
3. An inflation premium equal to the expected amount of inflation (i)

The Fisher Effect

- States that the nominal interest rate =
Real rate of return + expected inflation rate

Nominal rate of return

Percentage change in
wealth from an
investment (change in
number of dollars)

Real rate of return

Percentage change in
purchasing power from an
investment (change in what
you can buy with those
dollars)

Fisher Effect

$$(1 + r) = (1 + a)(1 + i)$$

Where

r = nominal rate of return

a = real rate of return

i = expected inflation rate

Useful approximation

$$r \approx a + i$$

The Fisher Effect

Example

- Consider an investment in a bond that returns 10% a year. 10% is a nominal rate that does not account for the fact that prices of goods might increase (*inflation*):
- Assume there is only one good, apples, which currently sell for \$1, but next year they will increase to \$1.01 (inflation of 1%).
- How many more apples will I be able to buy in one year if I invest in the bond?

Input	$t=0$	$t=1$
Bond (10% nominal return)	\$1,000	\$1,100
Apple (1% inflation)	\$1.00	\$1.01
Number of apples I can buy	1,000	1,089.11
Real return		8.911%

The Fisher Effect

Example

- Suppose you earn 6% per year in a savings account and inflation is 2% over the year.
- What is the *nominal* rate of return on this investment?
- What is the *real* rate of return on this investment?
- If you invested \$1,000 in your account, what do the nominal and real rates of return mean?

Rearrange and solve for a , the real rate:

$$(1+r) = (1+a)(1+i) \Rightarrow a = \frac{(1+r)}{(1+i)} - 1 \Rightarrow a = \frac{(1.06)}{(1.02)} - 1 = 0.0392 = 3.92\%$$

- Your wealth will increase by \$60 (nominal).
- Your purchasing power will increase by only \$39.20 (real).

Generalized Fisher Effect

- Real returns are equalized across countries (h = home, f = foreign).

$$a_h = a_f$$

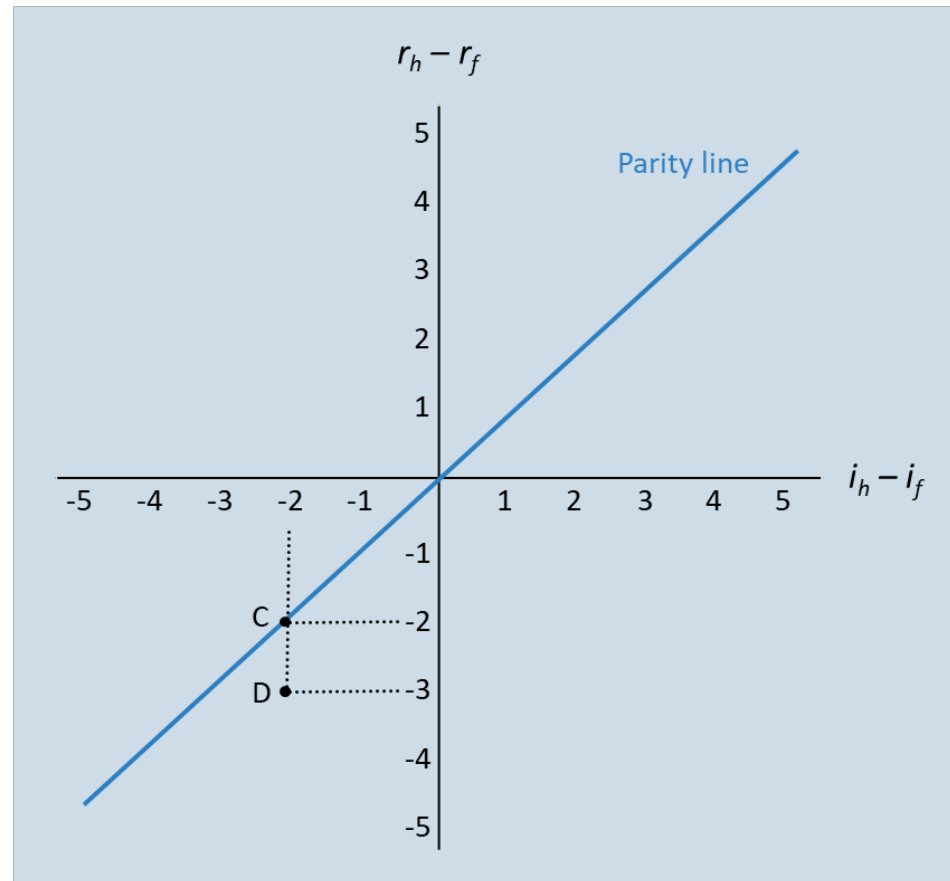
- Why? If real rates were higher in one country, capital would flow exclusively to it until the rates equalized.
- This means that: $\frac{1 + rh}{1 + ih} = \frac{1 + rf}{1 + if}$
- Or, rearranging: $\frac{1 + rh}{1 + r_f} = \frac{1 + i_h}{1 + i_f}$
- That is, the nominal interest rate differential must equal the anticipated inflation differential.

Comments on the next graph

1. At point C, higher rate of inflation in the foreign country by 2% is just offset by the 25 lower home currency interest rate.
2. At point D, the real rate of return in the home country is 1% lower than in the foreign country (an inflation differentiation of 2% vs an interest At point D, the real rate of return in the home country is 1% lower than that in the foreign country (an inflation differential of 2% compared to an interest differential of 3%). As a result, funds should flow from the home country to the foreign currency to take advantage of the real differential. This flow will continue until expected real returns are again

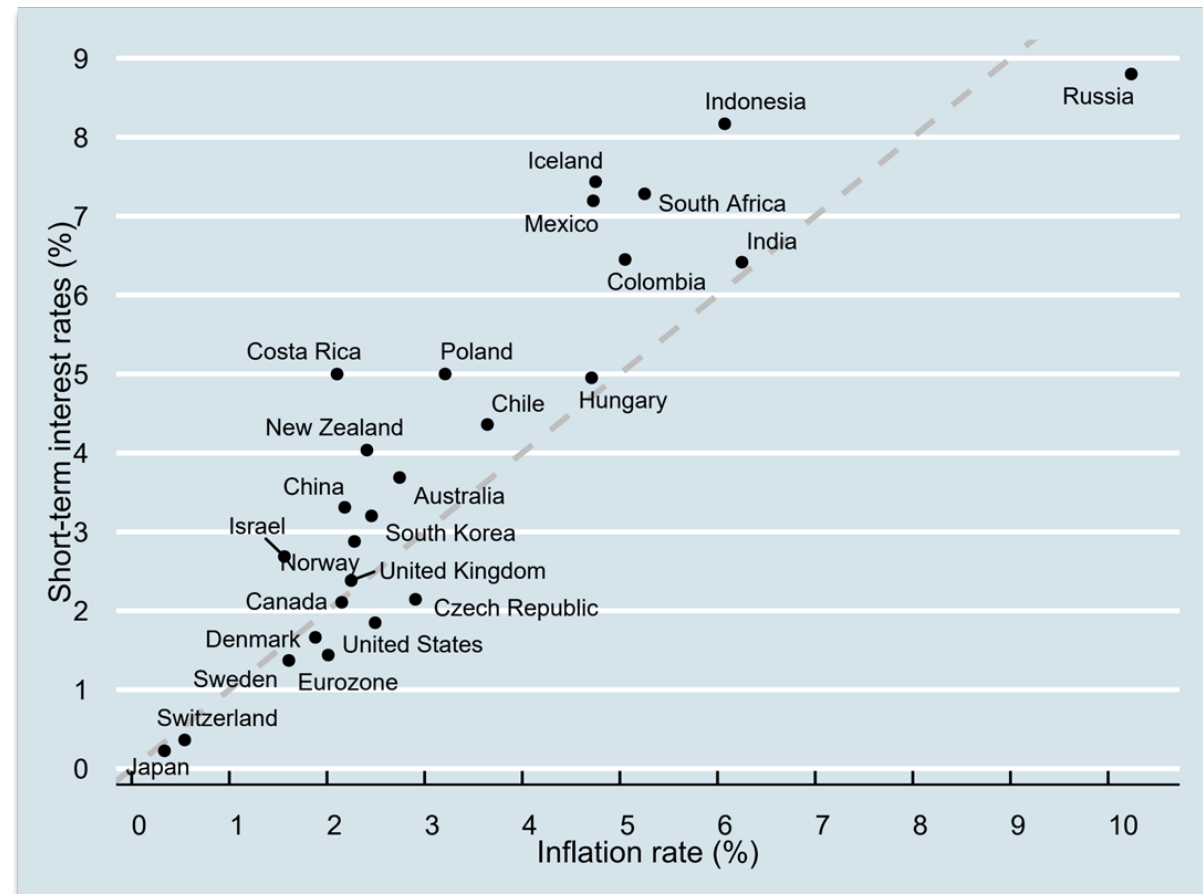
Intuitive Approximation of the Fisher Effect

$$\underbrace{r_h - r_f}_{\text{Difference in nominal interest rates}} \approx \underbrace{i_h - i_f}_{\text{Difference in inflation rates}}$$



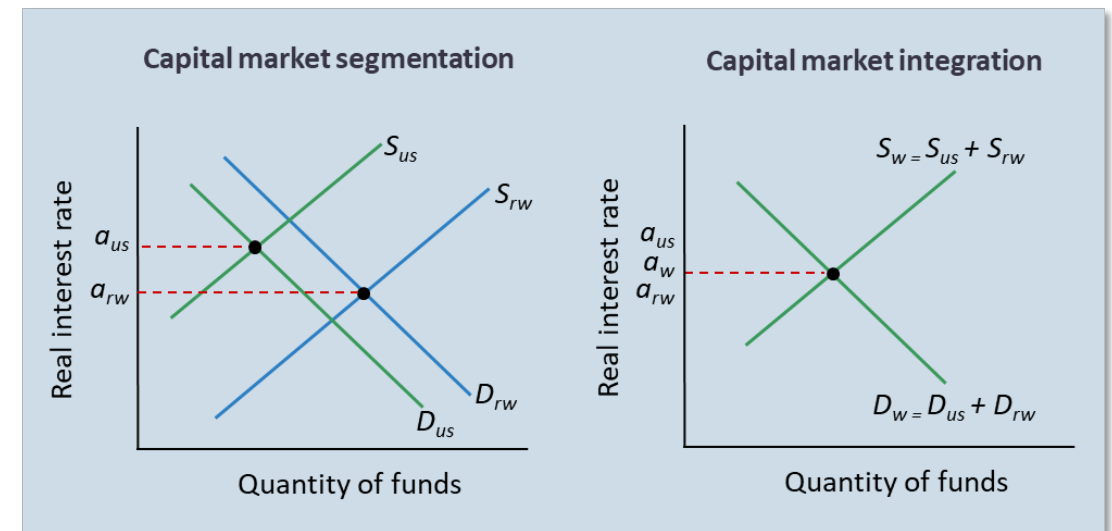
Empirical Evidence for the Fisher Effect

- Short-term nominal interest rates vs. inflation rate (2017)
- Countries with higher inflation have higher nominal rates.
- General consensus: differentials in real rates of return between countries are small.



Empirical Evidence for the Fisher Effect

- Empirical evidence is consistent with the hypothesis that most variation in nominal interest rates across countries is attributed to differences in inflationary expectations.
- Interest differentials could not survive long in internationalized capital markets.
 - **Capital market segmentation:**
 - The US real interest rate a_{us} is based on national demand D_{us} and national supply S_{us} of credit.
 - The real rate in the rest of the world a_{rw} is based on rest-of-world supply S_{rw} and demand D_{rw} .
 - **Capital market integration:**
 - The new world a_w is based on the rest-of-world supply $S_w (S_{us} + S_{rw})$ and demand $D_w (D_{us} + D_{rw})$.

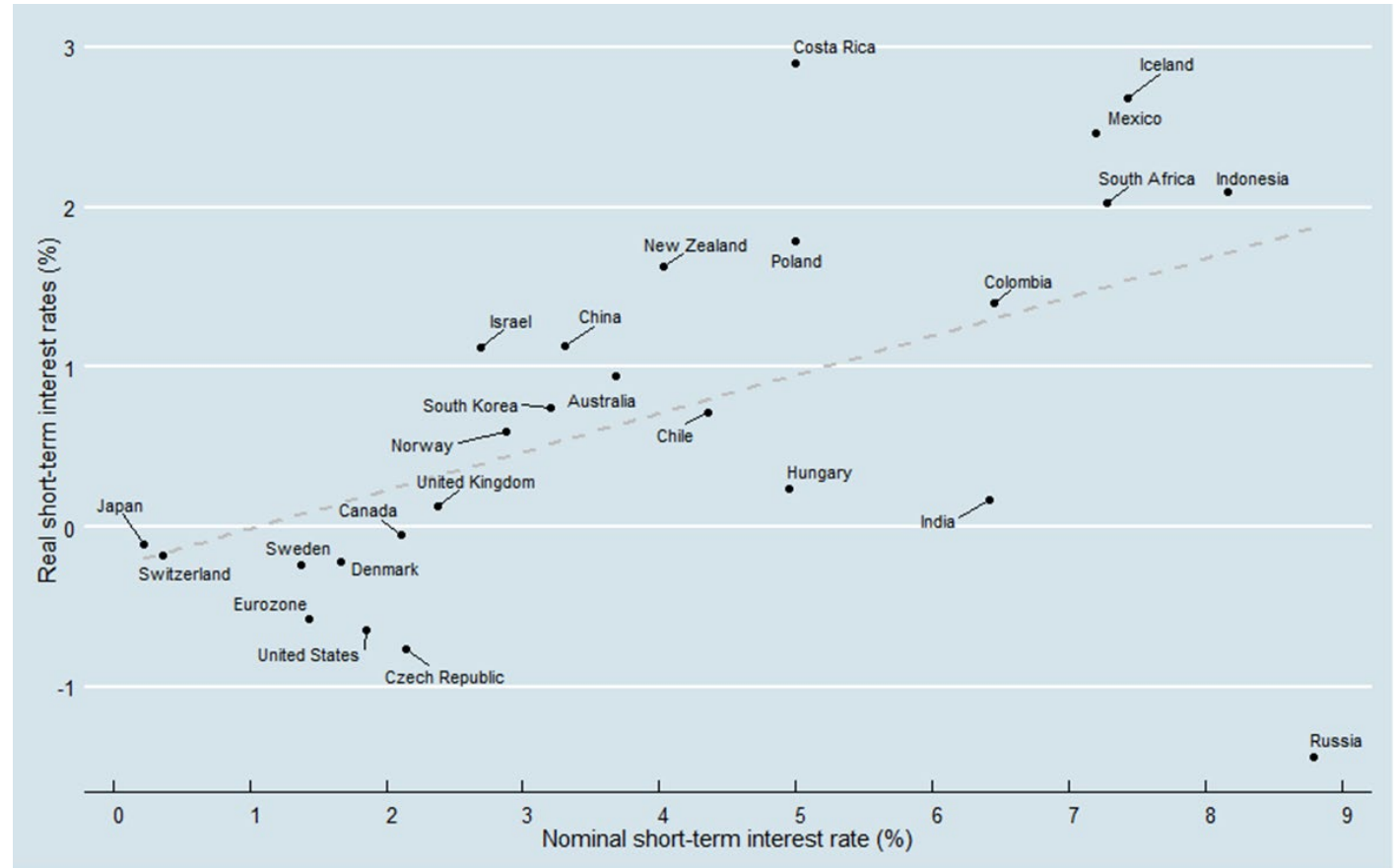


Financial Market Integration (comments for the last slides)

1. As in the US, the real interest rate is higher than that of the rest of the world. Once the market opens up, there will be a shift from the US credit market to the rest of the world. This indicates that the real interest rate in the US will fall while the real interest rate from the rest of the world will rise, ultimately leading to a new equilibrium in global/world real interest rates, as shown in the previous diagram.
2. Demand for credits globally will increase the real interest rate (ie in an integrated market).
3. However, real interest rate differential could persist, if investors prefer to hold onto their domestic assets and avoid exposure to exchange risk. The next graph demonstrates this phenomenon. They may hold to their own domestic assets and avoid exposure to exchange risk. Next graph is a demonstration of this phenomenon

Empirical Evidence for the Fisher Effect and Market Integration

- Real interest rates versus nominal interest rates for 25 developed and developing countries plus the Eurozone (2021).
- In developing countries, there are greater expectations for higher inflation; therefore, real short-term interest rates tend to be higher. Currency controls and other measures in these nations may impose political risks on foreign investors, who will seek higher interest rates.
- For example, if political risk in Brazil leads foreign investors to demand a 7% higher interest rate than what they would find elsewhere, these investors might consider a 10% expected real return in Brazil equivalent to a 3% expected real return in the US. In other words, real interest rates in developing countries may exceed those in developed nations, yet such investments may still not appear attractive.



International Fisher Effect (IFE)

- International Fisher effect
- An intuitive approximation of the IFE
- Does the IFE work?

International Fisher Effect (IFE)

- Is similar to relative PPP but uses interest rates to explain exchange rates instead of inflation rates.
- Remember that
 1. PPP states that a currency with higher inflation will decrease in value, and
 2. The FE states that currencies with higher inflation will have higher nominal rates.

Combining (1) and (2) gives us the IFE:
Currencies with higher nominal interest rates will decrease in value.

The international Fisher Effect formula is able to estimate “*the expected future exchange rate*”, as shown below

$$\bar{e}_t = e_0 \times \frac{(1 + r_h)^t}{(1 + r_f)^t}$$

International Fisher Effect

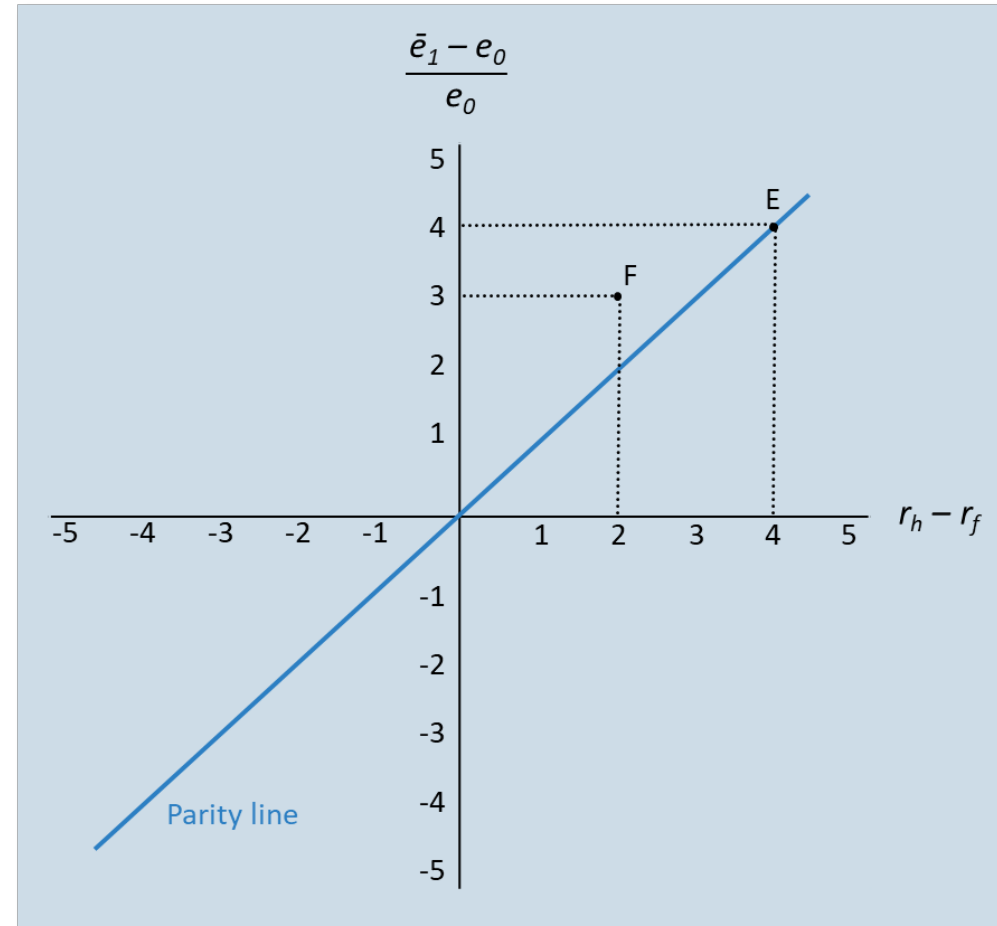
Example

- Suppose interest rates are expected to be 8% in the UK and 6% in the US.
- If the current exchange rate is \$1.6545/£, what is the expected exchange rate in one year if IFE holds?

$$\bar{e}_t = e_o \times \frac{(1 + r_h)^t}{(1 + r_f)^t} \rightarrow$$

An Intuitive Approximation of the IFE

$$\underbrace{\frac{\bar{e}_1 - e_0}{e_0}}_{\text{Percentage change in exchange rate}} \approx \underbrace{r_h - r_f}_{\text{Difference in nominal interest rates}}$$



Comments on the previous graph.

1. Point E represents a position of equilibrium because it is located on the parity line, with the 4% interest differential favouring the home country being offset by the expected 4% appreciation.
2. Point F shows disequilibrium. If the foreign currency is expected to appreciate by 3% in terms of HC, but the interest rate differential in favour of the home currency is only 2%, then funds would flow from the home country to the foreign country to take advantage of the higher exchange-adjusted return there.

Interest Rate Parity (IRP)

- Interest rate parity
- Arbitrage example
- No-arbitrage condition
- An intuitive approximation of IRP
- Empirical evidence of IRP

Forward premium vs Forward discount

1. A "forward premium" occurs when the forward exchange rate for a currency is higher than its current spot rate, meaning the market expects the currency to appreciate in the future,
2. while a "forward discount" happens when the forward rate is lower than the spot rate, indicating an expected depreciation of the currency;
3. for example, if the spot rate for **Euros is \$1.10 and the 90-day forward rate is \$1.12**, this represents a forward premium on the Euro, while a rate **of \$1.08 would be** a forward discount.
- 4.

Interest Rate Parity

The Theory states:

The forward rate differs from the spot rate at equilibrium by an amount equal to the interest differential ($r_h - r_f$) between two countries.

The forward premium or discount equals the interest rate differential between two countries.

Interest Rate Parity

- Interest rate parity ties the spot rate with the forward rate.

Spot rate

Today's exchange rate
for **immediate** delivery

Forward rate

Today's exchange rate
for **future** delivery

- Example: On April 29, 2020,
 - The spot rate was \$1.0873/EUR.
 - The 3-month forward rate was \$1.0900/EUR.

So, in three months, you can buy 1 euro for \$1.0900.

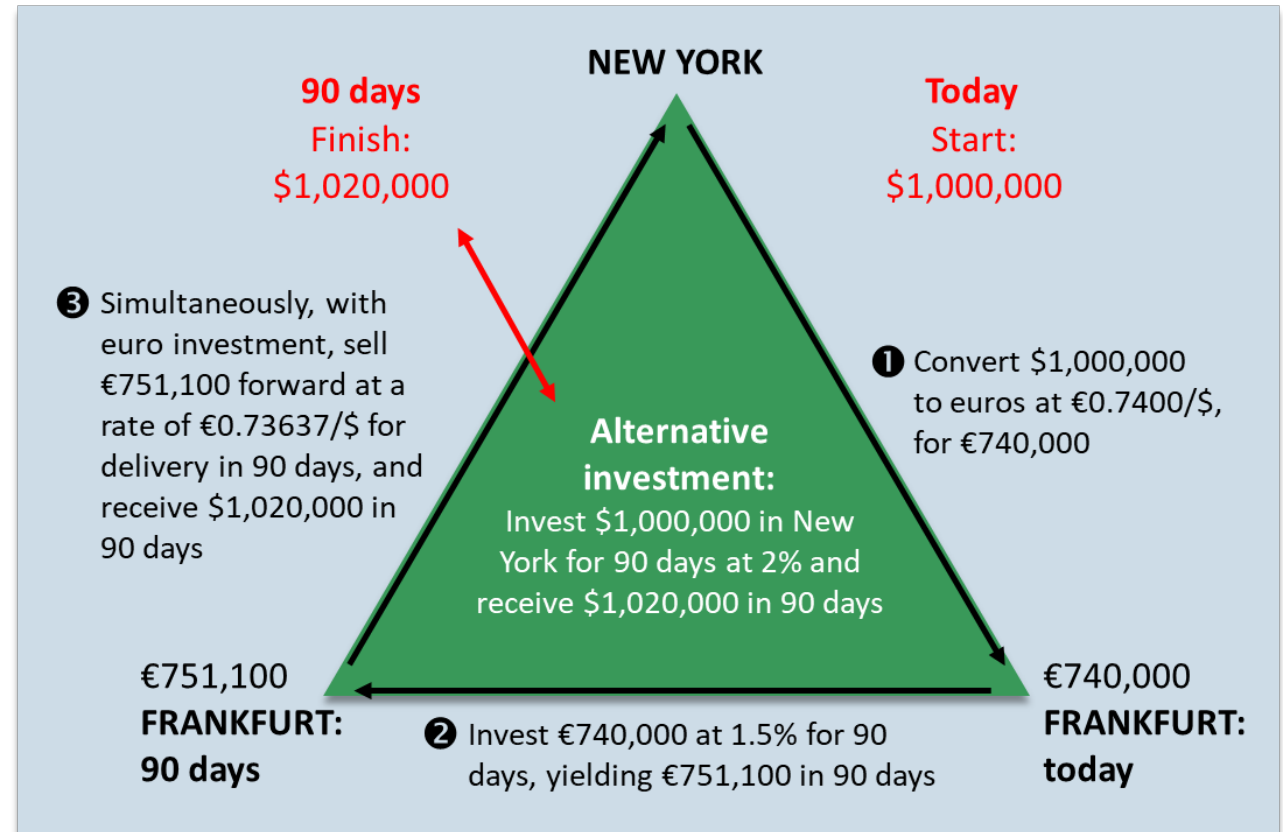
Interest Rate Parity

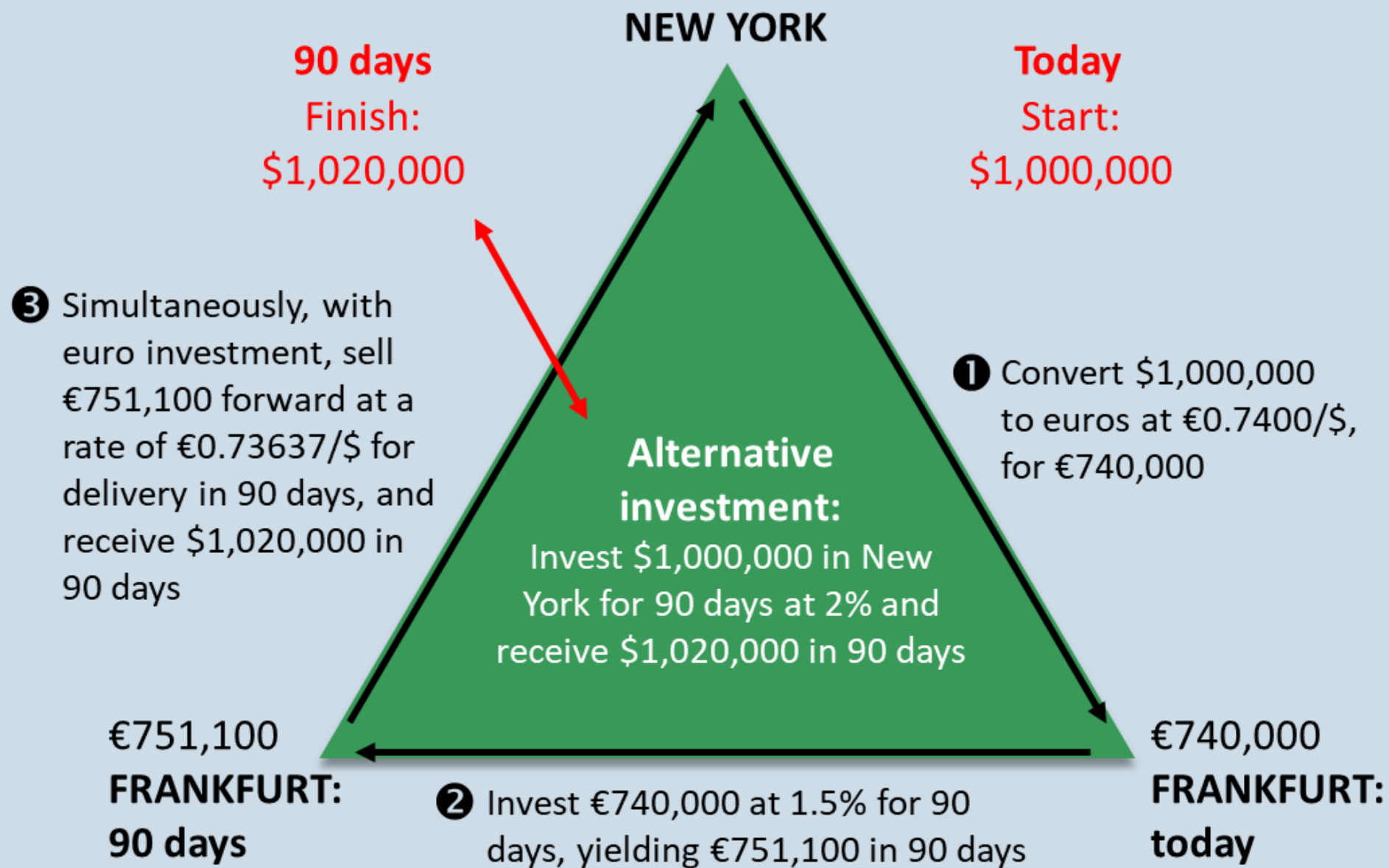
- The forward rate premium or discount to the spot rate is related to the **interest rate differential through arbitrage.**
 - *Forward premium* – expected future price for a currency $>$ spot price
 - *Forward discount* – expected future price for a currency $<$ spot price

Illustration of Interest Rate Parity when it applies

- Suppose
 - Spot rate is €0.74000/\$
 - 3-month forward rate is €0.73637
- Consider the following choices for investing \$1 million:
 - Invest in the US at 8% per annum for 3-months (2% for 3 months)
 - Or invest in the Eurozone at 6% per annum for 3-months (1.5% for 3 months)

Both strategies return the same!





No-Arbitrage Condition

- Funds will flow from the home to the foreign country, *if and only if*: $1 + rh < (1 + rf) \times \frac{f_1}{e_0}$
- The converse will happen if: $1 + rh > (1 + rf) \times \frac{f_1}{e_0}$
- The no-arbitrage condition is: $\frac{f_1}{e_0} = \frac{1 + rh}{1 + r_f}$

No-Arbitrage Condition

Example

- The US interest rate is 2.25% (per annum)
- The Japanese interest rate is 0.1%.
- The spot rate for the yen is \$0.009142.
- If interest rate parity holds, what is the 90-day forward rate?
- ***The example below shows that the Japanese yen should be selling at a premium.***

$$f_{90} = e_0 \frac{1 + rh}{1 + r_f} \Rightarrow f_{90} = 0.009142 \times \frac{1 + \frac{0.0225}{4}}{1 + \frac{0.001}{4}} = 0.009191$$

An Intuitive Approximation of IRP

$$\frac{f_1 - e_0}{e_0} \approx r_h - r_f$$

Forward
premium
or discount

Difference
in nominal
interest rates

Examples of Covered Interest Arbitrage

If the covered interest differential between two countries is non-zero, there is an arbitrage incentive to move money from one market to the other.

This movement of money to take advantage of a covered interest differential is known as **Covered Interest Arbitrage**.

Covered Interest Arbitrage

1. Conditions required:

- interest rate differential does not equal the forward premium or discount

2. Funds will move to a country with a more attractive rate.

New York

One year

Today

Finish:

7. Net profit equals
\$4,051.28.

6. Repay the loan plus interest
of \$1,070,000 out of
the \$1,074,051.28.

5. Collect the £574,358.97 and deliver
it to the bank's foreign exchange
department in return for
\$1,074,051.28.

London: one year

Start:

1. Borrow \$1,000,000 at an interest
rate of 7%, owing \$1,070.00
at year end.

2. Convert the \$1,000,000 to
pounds at \$1.95/£ for
£512,820.51.

3. Invest the £512,820.51 in
London at 12%, generating
£574,358.97 by year end.

London: today

4. Simultaneously, sell the £574,358.97 in
principal plus interest forward at a rate
of \$1.87/£ for delivery in one year,
yielding \$1,074,051.28 at year end.

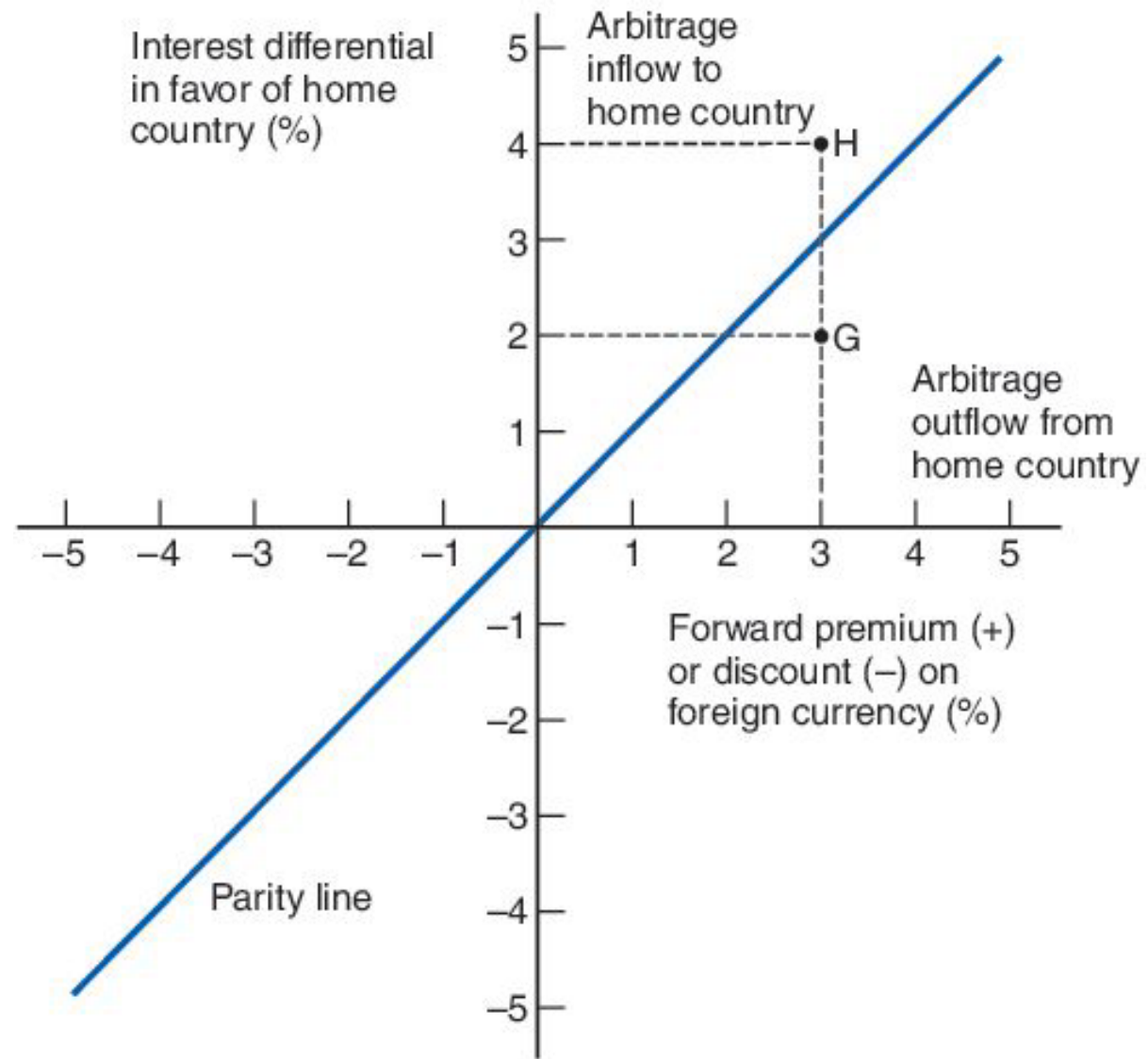
Interest Rate Parity

Market pressures develop: (based on the previous example and slide)

- a. When one currency is in higher demand in the spot market and sold forward, this will lead to an increase in the spot price and a decrease in the forward price.
- b. An inflow of funds tends to lower interest rates; specifically, an inflow of money into the UK reduces the interest rate there. At the same time, as more individuals exchange US dollars for Pounds, interest rates in the US will rise.
- c. Parity is eventually reached.

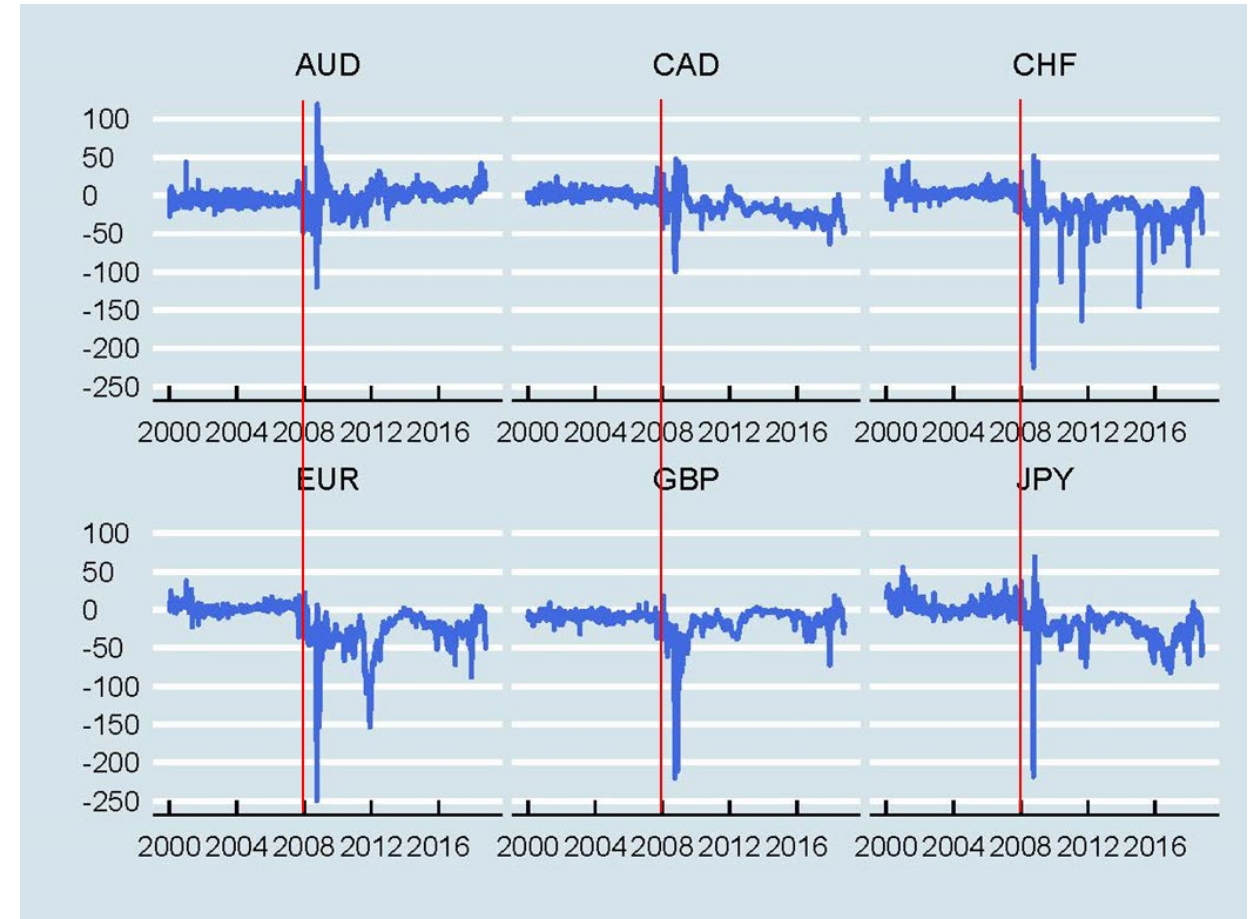
Summary of the forces shown in the next slide

1. Point G indicates a situation of disequilibrium. Here, the interest differential is 2%, whereas the forward premium on the foreign currency is 3%. Transferring funds abroad with the exchange risk covered will yield an additional 1% annually. At point H, the forward premium remains at 3%, but the interest differential increases to 4%. Now, reversing the flow of funds becomes profitable. The 4% higher interest rate more than compensates for the 3% loss on the forward exchange transaction, resulting in a 1% increase in the interest yield.



Empirical Evidence of IRP

- Deviations from covered interest rate parity were very small prior to the great financial crisis (2008-2010).
- During the crisis, there are large deviations as arbitrageurs face a lack of funding.
- But why do large deviations exist after the crisis?
 - Research points to new regulation preventing arbitrage from taking place, especially at the end of the quarter when banks disclose their capital requirements.



Unbiased Forward Rates (UFR)

- How unbiased is the UFR?
- Relationship between the forward rate and future spot rate

How Unbiased is the UFR?

- Consider the IFE and IRP

IFE

$$\bar{e}_t = e_0 \frac{1+rh}{1+r_f}$$

IRP

$$f_1 = e_0 \frac{1+rh}{1+r_f}$$

- Note the right hand side of each equation is equal: $f_1 = \bar{e}_t$

- However, evidence suggests that
 - The forward rate is not unbiased; and
 - There is a risk premium:

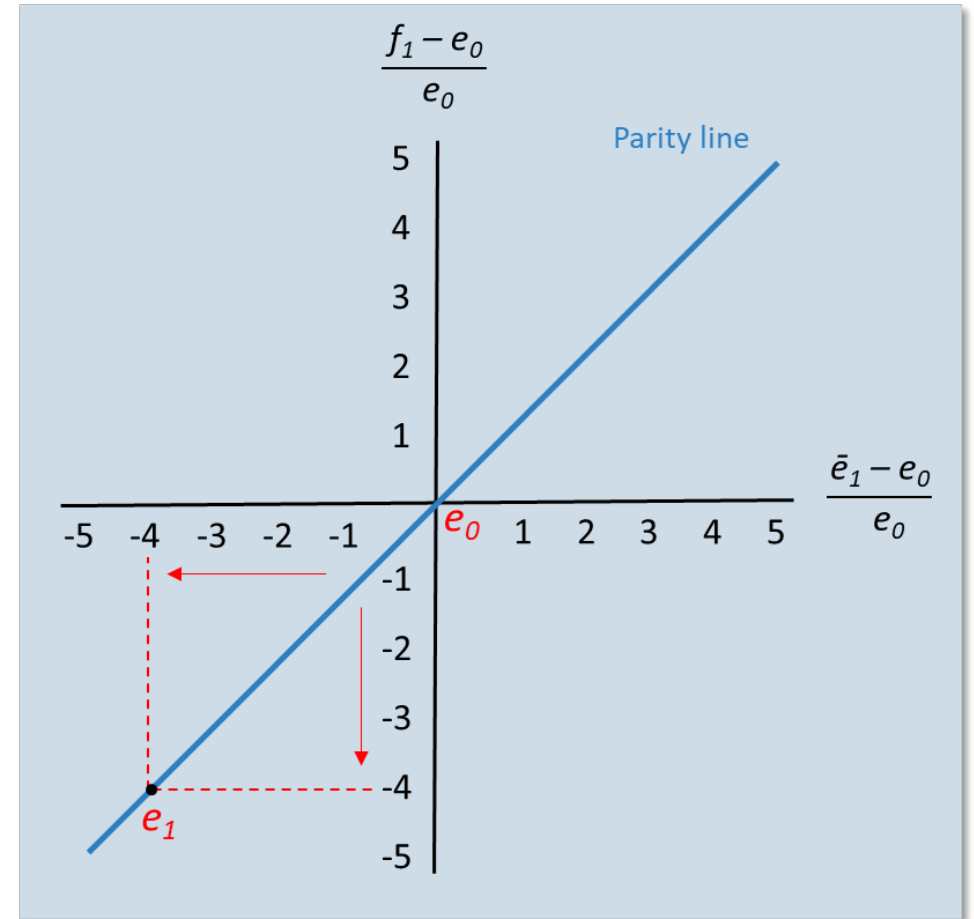
$$f_1 = \bar{e}_t + RP$$

Where RP is risk premium, which can be positive or negative

Relationship Between the Forward Rate and Future Spot Rate

Example

1. At equilibrium $e_0, \bar{e}_0 = f_1$ and there is no incentive to buy or sell currency forward.
2. The value of the pound is expected to depreciate by 4%.
3. Pound holders sell pounds forward, dollar holders delay converting dollars to pounds.
4. f_1 decreases over time by 4% to establish a new equilibrium at e_1 .



Currency Forecasting

7.1 Requirements for successful currency forecasting

7.2 Market-based forecasts

7.3 Forecasting in a fixed-rate system

7.4 Model-based forecasts

7.5 Model evaluation

Requirements for Successful Currency Forecasting

Exclusive use of
superior
forecasting
model

Consistent early
access to
information

Exploits small,
temporary
deviations from
equilibrium

Can predict
government
intervention in
forex market

Not sustainable as
imitators enter market

Not sustainable with
instant information

Deviations do
not last long

Long-run viability if it is not
necessary to outguess the
market to win

Market-Based Forecasts

Forward rates

- Derived most simply from current forward rates
- f_1 serves as an unbiased estimate of the spot rate as of that date
- Forecasting limited to one year due to general absence of long-term forward contracts

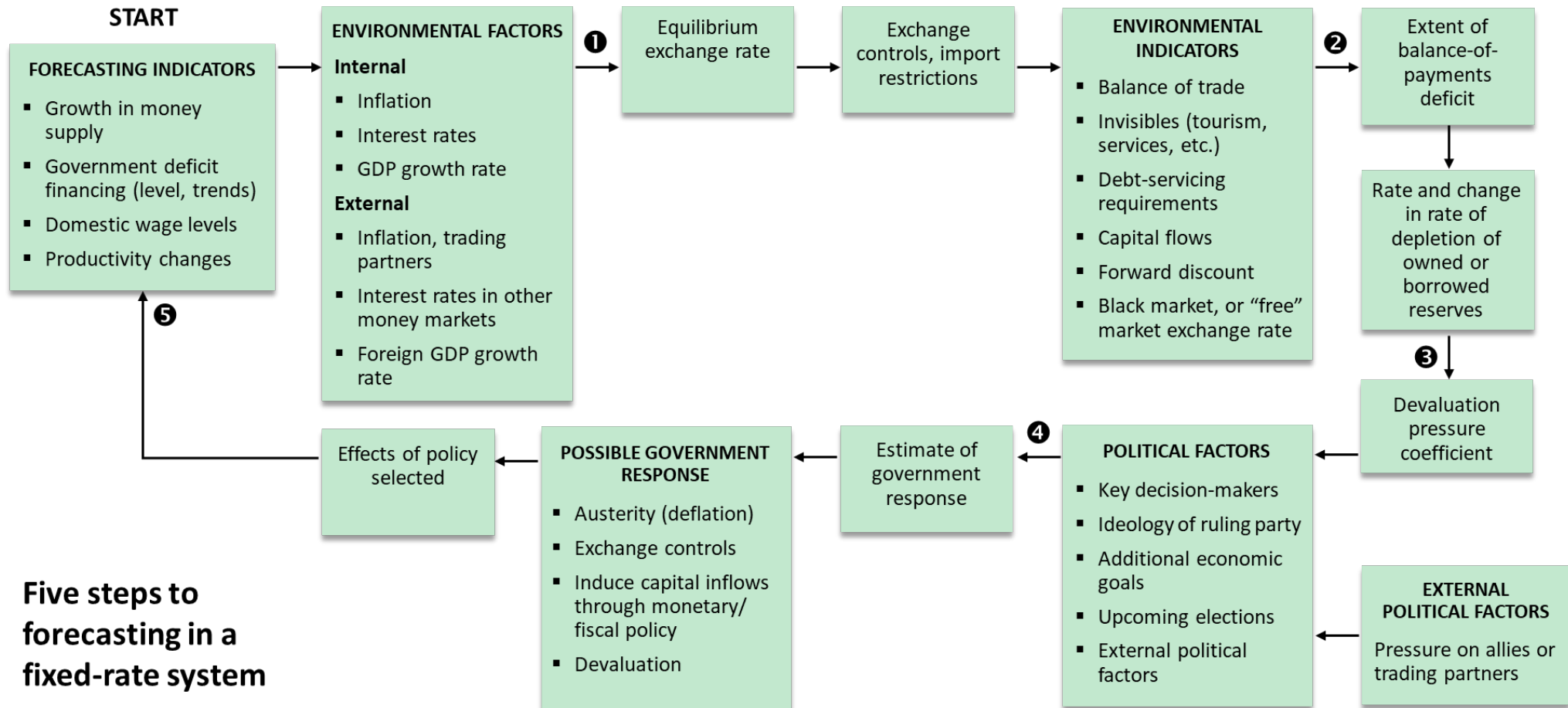
Interest rates

- Interest rate differentials can be used to forecast beyond one year.
- Example:
 - 5-year \$ interest rate = 6%
 - 5-year € interest rate = 5%
 - Current \$/€ = \$0.90
 - Value of € in 5 years = e_5

If \$1.00 is invested in euros today, what will it be worth in 5 years? Solve for e_5 :

$$\frac{(1.05)^5 \times e_5}{0.90} = (1.06)^5$$

Forecasting in a Fixed-Rate System



Model-Based Forecasts

- Fundamental analysis
 - Most common approach
 - Relies on painstaking examination of macroeconomic variables and policies that may include a currency's movements.
 - Relative inflation
 - Interest rates
 - National income growth
 - Changes in money supplies
 - Simplest analysis uses PPP

Example

- Average expected annual US inflation = 2.1%
- Average expected annual South African inflation = 4.9%
- If the current spot rate for the rand is \$0.0690, what is e_2 ?

Using PPP:

$$\$0.0690 \times (1.021 / 1.049)^2 = \$0.0653$$

Model-Based Forecasts

- Technical analysis
 - Focuses exclusively on past price and volume movements.
 - Ignores economic and political factors.
 - Success depends on discovering price patterns.

Two types of technical analysis

Charting

Using bar charts and sophisticated computer-based extrapolation techniques to find price patterns

Trend analysis

Using mathematical computations to identify price trends

Model Evaluation

- Profitably forecasting exchange rates is inconsistent with the efficient market hypothesis.
- Exchange rates fluctuate randomly as market participants assess and react to new information.
- Thus, exchange rate movements are unpredictable, and arbitrage profits cannot persist.
- Forecasting criteria
 - **Accuracy** – focuses on deviations between the actual and forecasted rates
 - **Correctness** – Examines whether the forecast predicts the right direction of the change in exchange rates
 - *An accurate forecast may not be correct in predicting the direction of change, and a correct forecast may not be very accurate.*

Decision rule for deciding whether to use forecasts to hedge with forward contracts

If $f_1 > e_1$, sell forward

If $f_1 < e_1$, buy forward

Calculating profit (loss) percentage

When $f_1 > e_1 : 100 \times [(f_1 - e_1) / e_1]$

When $f_1 < e_1 : 100 \times [(e_1 - f_1) / e_1]$